Yoosun Kim

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Objective

Interested in imaging through scattering media and developing computational reconstruction methods that integrate optics, physics-based modeling, and deep learning

Education

- **Korea Advanced Institute of Science and Technology (KAIST)** *Aug 2024 - Present*
  M.S. in Bio & Brain Engineering Daejeon, Republic of Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** *Feb 2020 - Aug 2024*
  B.S. in Physics and Bio & Brain Engineering Daejeon, Republic of Korea
  - GPA: 3.70/4.30
  - Relevant Coursework: Optics, Laser Optics, Advanced Wave Optics, In Vivo Imaging System, Computational Physics, Computer Vision

Research Experience

- **Institute of Innovation for Future Army, Republic of Korea Army** *Jul 2025 - Present*
  Research Soldier (Active Duty Military Service) Daejeon, Republic of Korea
  - Contributed to statistical analysis of temporal scan data from cylindrical pipe interiors to identify degradation trends
- **Optical Imaging Lab (PI: Prof. Mooseok Jang)** *Sep 2024 - Present*
  M.S. Student (currently on leave for active duty military service) Daejeon, Republic of Korea
  - Enabled phase imaging enclosed by dynamic scattering layer using focused illumination and a physics-guided neural reconstruction framework (*Optica*)
  - Derived optical forward models for dynamic scattering media considering spatial coherence and optical memory effects (*IEEE TPAMI*)
  - Optimized SLM modulation patterns as trainable parameters for optical-field-based MNIST classification
- **Optical Imaging Lab (PI: Prof. Mooseok Jang)** *Jan 2022 - Aug 2024*
  Undergraduate Researcher Daejeon, Republic of Korea
  - Developed occluder-based passive non-line-of-sight imaging using supervised learning with a ray-optics forward model
  - Applied Bayesian optimization for hyperparameter selection in Wirtinger-based holographic reconstruction
  - Correlation imaging through scattering layers/around corners via speckle patterns
  - Implemented and evaluated phase retrieval methods including GS, HIO, Wirtinger Holography, and Deep Image Prior
  - Built a digital off-axis holography setup for phase imaging of red blood cells

Publications

J=Journal, C=Conference, S=Submitted/Under Review

[J.1] **Y. Kim***, **G. Song***, H. G. Kim, C. S. Lee, M. Jang (2026). **Single-shot Imaging of Phase Objects Fully Enclosed by Dynamic Scattering Layers**. *Optica*, Vol. 13, No. 7, pp. 1245–1253.

[J.2] T. Kwon*, G. Song*, **Y. Kim**, J. Kim, J. C. Ye, M. Jang (2025). **Video Diffusion Posterior Sampling for Seeing Beyond Dynamic Scattering Layers**. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 47, No. 12, pp. 11152–11167.

[C.1] **Y. Kim**, M. Jang (2024). **Reliable Reconstruction of Passive Non-Line-of-Sight Imaging with Occluder by Deep Learning**. In *SPIE Advanced Biophotonics Conference (SPIE ABC 2023)*, Vol. 13076. SPIE.

* denotes equal contribution

Other Projects

- **Biomedical Signal Processing & AI Projects**
  - Analyzed EMG, ECG, and EEG signals using frequency-domain filtering in LabVIEW
  - Designed and implemented band-pass filter circuits for PCG signal processing
  - Applied protein language models to estimate viral mutation risk
  - Performed MRI and CT image reconstruction using 2D Fourier domain filtering
- **Computational Imaging & Photonics Projects**
  - Implemented phase retrieval algorithms using pre-trained flow matching models
  - Designed and implemented a second-harmonic generation (SHG) setup based on an Nd:YAG laser to study homogeneous and inhomogeneous spectral broadening
  - Designed, built, and experimentally characterized a laser cavity

Skills

- **Programming:** Python, MATLAB, PyTorch
- **Hardware:**
  - Optical Setups
  - Spatial Light Modulator (SLM), CMOS/CCD camera systems
  - LabVIEW
- **Language:** Korean (Native), English (Fluent)

Honors and Awards

- **Best Poster Award** *Aug 2024*
  4th Quantitative Phase Imaging Symposium (KAIST)
  - Imaging phase objects between two dynamic scattering media with an untrained neural network (Selected for Top 5 oral presentations among 30 posters)
- **ASAN Biomedical Science Undergraduate Scholarship** *Feb 2024*
- **Young Investigator Award** *Nov 2023*
  SPIE Advanced Biophotonics Conference
  - Reliable reconstruction of passive non-line-of-sight imaging with occluder by deep learning (Selected as a Top 10 oral presentation)
- **Loonshot Startup Challenge** *Sep 2022*
  KAIST
  - A data-driven relationship visualization platform (Team Leader)
- **KAIST Alumni Academic Scholarship** *2022 - 2024*

Teaching and Leadership

- **Teaching Assistant:**
  - **Principles of Biomedical Optics**: Designing bright and dark-field microscopy experiments
  - **Instrumental Bioengineering**: Building a Mach-Zehnder interferometer and conducting digital off-axis holography experiments
  - **Computational Biomedical Optics**: Phase retrieval with the GS algorithm, and implementation of the angular spectrum method
- **Tutor:** KAIST Cyber Talented Education Center Tutor (Python), (Feb 2022 - Nov 2022)
- **Campus Activity:**
  - Member, KAIST Department of Physics undergraduate student council (Mar 2021 – Dec 2021)
  - Member, KAIST Festival Preparation Committee (Mar 2020 – Dec 2021)